



Mapping Urban Heat & Environmental Resilience with Science Gateways from the Bluff to the Bayou



[Presenter Name] · [Institution] — [Course Prefix & Number] — FacultyHack@Gateways 2026

ABSTRACT

UFOR 271 Environmental Science is a gateway- and AI-enhanced Course-Based Undergraduate Research Experience (CURE) that engages undergraduates in investigating environmental resilience across South Louisiana communities. Students move through a four-stage research arc data discovery, visualization, analysis, and interpretation using browser-based Jupyter notebooks hosted on ACCESS CloudBank Classroom, together with open science gateways (OpenTopography, iNaturalist, EPA EnviroAtlas) and interactive mapping tools (geemap / leafmap). Artificial intelligence is integrated deliberately and transparently: a tiered AI-use policy (prohibited / limited / permitted) matched to specific tasks, mandatory disclosure and verification, and summative assessments redesigned around authentic process to preserve integrity. High-performance computing is introduced at an introductory, hands-on level only where the data genuinely warrants it, keeping the focus on student learning rather than technology. The course removes local-setup barriers, ensures equitable access for all students at this Historically Black University, and builds transferable data-literacy, computational-thinking, and responsible-AI skills while addressing real community and environmental-justice questions. Southern University and A&M College requests CloudBank Classroom to support UFOR 271 Environmental Science (Fall/Spring). The course engages undergraduates in a semester-long Course-based Undergraduate Research Experience investigating environmental resilience in South Louisiana communities. Using browser-based Jupyter notebooks, 10–15 students will access authentic geospatial and environmental datasets to visualize, analyze, and interpret local environmental patterns, building data literacy and computational skills while addressing real community and environmental-justice questions. The managed environment removes local-setup barriers and ensures equitable access for students at this HBCU..

INTRODUCTION

UFOR 271: Environmental Science connects foundational environmental science with authentic South Louisiana environmental challenges. The redesign integrates browser-based science gateways, reproducible Jupyter notebook workflows, geospatial data, and a collaborative Course-based Undergraduate Research Experience (CURE). Computational tools are used selectively where they deepen investigation and analysis while minimizing unnecessary technical barrier.

Course Goals

- Disciplinary:** Collect, document, manage, visualize, and interpret environmental field and secondary data.
- Skill Development:** Design and conduct collaborative environmental research using responsible and reproducible practices.
- Professional Practice:** Evaluate environmental hazards and apply environmental ethics, economics, laws, policies, and equity considerations to authentic decisions.
- Gateway/HPC/AI Goal:** Use browser-based science gateways and Jupyter notebooks, develop foundational HPC vocabulary, and apply AI responsibly through verification, disclosure, and appropriate limitations.

Grade Computation

- Formative Assessments : 200 points (40%):** OER reading checks; environmental field journal; four environmental-data/gateway activities; five CURE milestones; participation.
- Summative Assessments — 300 points (60%):** Environmental Systems Exam (50); Decision-Making Case + AI-use appendix (50); CURE Research Report (75); Scientific Poster & Presentation (50); Community Environmental Product (25); Final Environmental Solutions Assessment (50).
- Grading Scale:** A = 450–500 • B = 400–449 • C = 350–399 • D = 300–349 • F = below 300.

Mentor Feedback Incorporated

- CloudBank JupyterHub Integration:** ACCESS CloudBank Classroom was designated as the managed, browser-based JupyterHub environment to reduce local software-installation barriers and provide a consistent computational environment.
- Scaffolded CURE Workflow & Fallbacks:** The CURE uses four research stages Discovery, Visualization, Analysis, and Interpretation with spreadsheet/CSV alternatives so students can achieve the same learning outcomes when gateway access is limited.
- HPC Carpentry On-Ramp:** An introductory HPC Carpentry experience is incorporated during Weeks 1–2 to establish vocabulary and familiarity with remote computing without requiring advanced parallel programming.

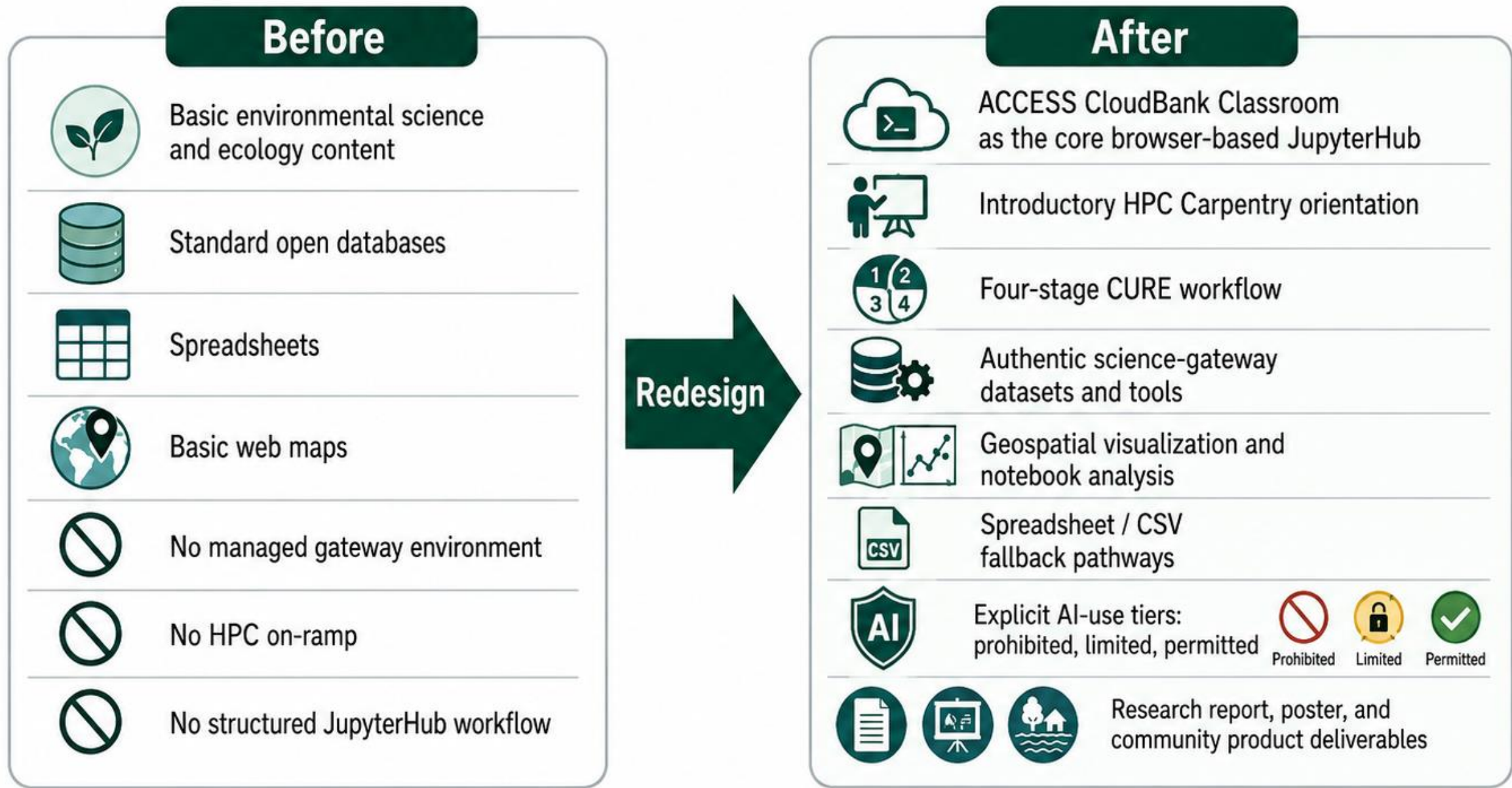
IMPACT ON STUDENTS

Democratize access to HPC services for students at HBCU. User Population: 10–15 undergraduate Urban Forestry students at Southern University and A&M College. Designed Impact: Reduce software and setup barriers through browser-based, preconfigured computational tools. Provide structured access to authentic environmental and geospatial datasets. Build experience with data discovery, visualization, analysis, interpretation, and reproducibility. Introduce HPC concepts without requiring advanced programming. Develop responsible AI practice through explicit limits, verification, disclosure, and accountability. Maintain access through spreadsheet/CSV fallback pathways.

COURSE REDESIGN: BEFORE & AFTER

Course Redesign: Before → After

Environmental Science course transformation



AUTHORS & MENTORS



Tanganika Johnson, Ph.D.
Faculty Participant
Southern University and A&M College
tanganika.johnson@sus.edu



Mohammed Elmellouki, Ph.D.
Faculty Mentor
Mississippi Valley State University
mohammed.Elmellouki@mvsu.edu

PROJECT GITHUB REPO

<https://github.com/tanganjohnson-ai/Hackathon-26-Environmental-Science.git>

Sponsors: SGX3 · National Science Foundation · ORNL · Omnibond · TACC · HackHPC

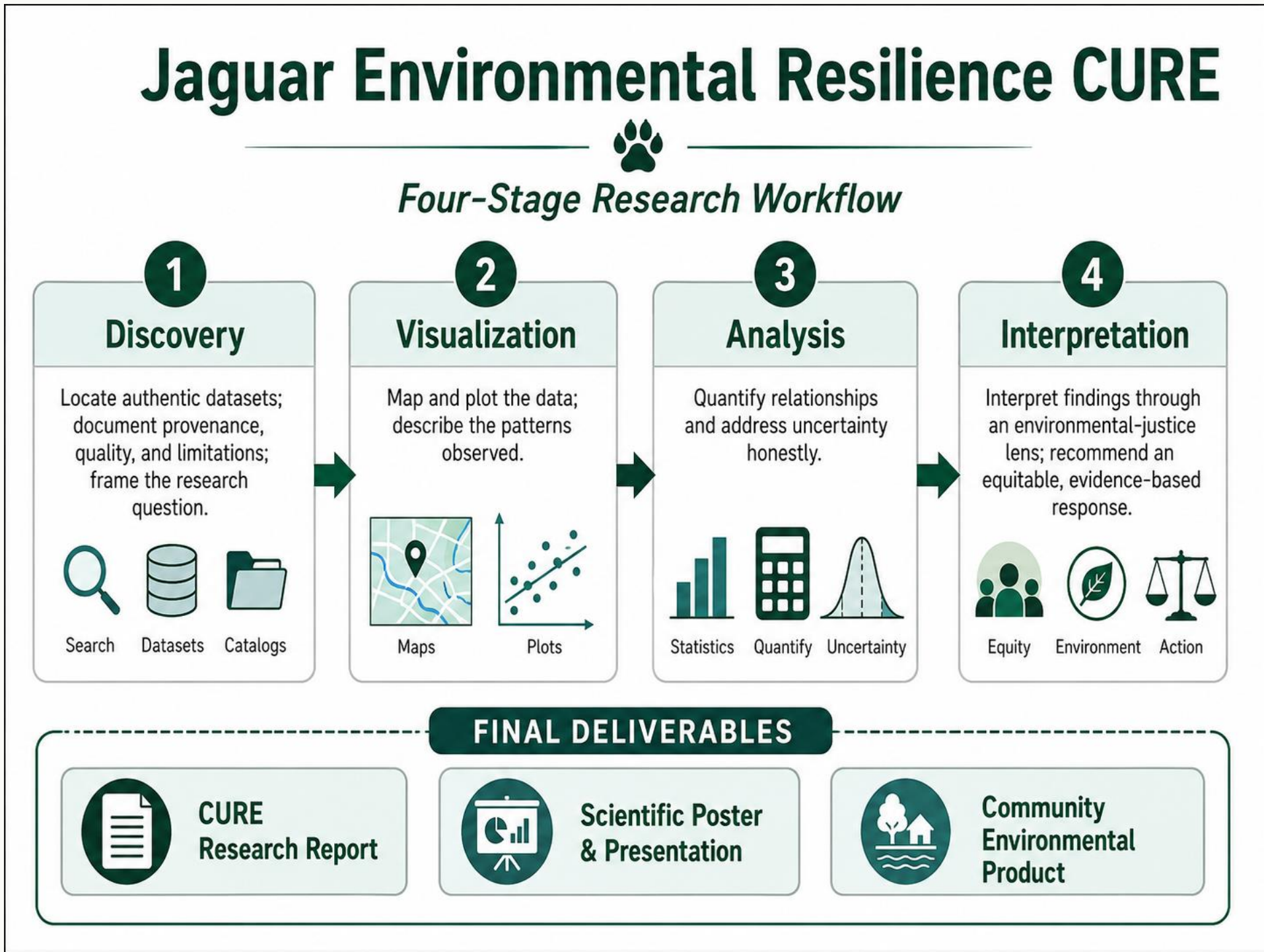


FacultyHack @Gateways26

hackhpc.github.io/facultyhack-gateways26



DELIVERABLE EXAMPLE



Project Goals

- Collect, document, manage, visualize, and interpret authentic environmental data.
- Conduct collaborative environmental research using responsible and reproducible practices.
- Evaluate evidence, uncertainty, environmental risk, and equity.
- Communicate findings to scientific, professional, and community audiences.
- Propose an evidence-based and equitable response to a South Louisiana environmental challenge.

EXPANSION IDEAS

New Topics & Datasets: Extend the CURE framework to additional environmental-resilience questions and datasets, including soil contamination.

Follow-on Opportunities: Introduce advanced geospatial tools such as Google Earth Engine/geemap and connect the experience to upper-level Urban Forestry research electives.

Community Partnerships: Develop partnerships with Baton Rouge resilience organizations and municipal planning agencies to strengthen public dissemination of student environmental products.

REFERENCES

Introduction to Environmental Science, 2nd Edition. Open Textbook Library.
HPC Carpentry. *Introduction to High-Performance Computing*.
ACCESS Cloud Bank Classroom.
OpenTopography.
EPA EnviroAtlas.
iNaturalist.
geemap / leafmap.
HazMapper (DesignSafe).
ScienceGateways.org.

Course Redesign Integration

Gateway, HPC, CURE, AI, and assessment connections in Environmental Science

